



اونيورسيتي مليسيا فهغ السلطان عبدالله
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Teaching Factory UPTVET

TECHNICAL EVIDENCE REPORT
IR4.0 PRODUCTION LINE REPAIR USING NEW WIRING TECHNOLOGY APPROACH

1.0 OVERVIEW

I'm involved on Student Work Scheme on Problem-Based Learning project in a 3 person in a team supervised by my lecturer which is Ir. Ts. Mohamad Rahimi Bin Mohamed Rodzi and Ir. Ts. Nik Mohd Zaitul Akmal Bin Mustapha. This project involved the comprehensive restoration and upgrading of an Car Absorber Assembly Production Line in the Teaching Factory UMPSA. My scope was focused on upgrading Workstations 2 and 3 from a legacy state to a fully integrated, communicative system. This included a total overhaul of the electrical wiring, control logic (PLC/HMI), and the establishment of a communication network involving the Sub-Control Panel (SCP), Main Control Panel (MCP), and Collaborative Robots (Cobots).

2.0 OBJECTIVE

- To restore hardware functionality through systematic troubleshooting and modern rewiring techniques.
- To optimize station control by developing advanced PLC and HMI programs.
- To establish a unified communication network between the MCP, SCP, and COBOT for synchronized production.
- To ensure all physical and logical changes are documented accurately in the EPLAN system.

3.0 TASK & RESPONSIBILITY

3.1 Electrical Hardware & Troubleshooting

- **I/O Checklist & Verification:** Conducted a full I/O mapping to identify faulty sensors and actuators.
- **Dismantling Legacy Components:** Removed obsolete sensors and damaged wiring to prepare for the upgrade.
- **Modern Rewiring (SVB/Distributor Box):** Instead of traditional point-to-point wiring, I implemented a **Sensor-Actuator Distributor Box (SVB)** approach. This reduced cable clutter, simplified troubleshooting, and followed modern industrial "plug-and-play" standards.
- **VFD Installation:** Performed wiring for the Variable Frequency Drive (VFD) and configured specific **Function Codes** (Analog Input, Frequency Parameter, Three Line Control) to match system sequences.
- **Schematic Drawing Update:** Modifying the terminal connection of the sensor and valve, updating the PC I/O connection and revising the Bill of Material.

3.2 Programming & System Integration

- **Workstation PLC Programming:** Developed the sequence logic for Workstations 2 (Cap Insert station) and 3 (Insert Colar) to handle car absorber assembly process.
- **Inter-PLC & Cobot Communication:** Wrote the communication logic to link the **Sub** and **Main Control Panel**, VFD parameter and direction. I also integrated the **Cobot** into the sequence, ensuring it receives "Ready" signals before commencing its task.

- **HMI Programming:** Develop the supporting ladder logic to allow operators to monitor station status and toggle manual/auto modes including VFD parameter setting and control.
- **PLC Addressing:** Systematically updated the PLC address list to reflect the new I/O layout, ensuring no address conflicts in the global tag list.

3.3 Mechanical Installation

- Fabricated and installed custom **brackets** for Distributor Box and the HMI display.
- Installed new **wire ducts** for sensor and valve to ensure all cables were protected and organized according to safety standards.

4.0 TECHNICAL RESOURCES

4.1 Eplan Report

Here is the technical drawing report of the system:

https://drive.google.com/file/d/16H6DhpgPNT0322DpyYoKmgzk6_iE0MFi/view?usp=sharing

4.2 Project Work Schedule & Documentation

A summary sheet and work schedule were maintained to track progress against milestones.

<https://docs.google.com/spreadsheets/d/1TfPJfEB3IrhCkh-7O7vTHGRgPIo4nMQp/edit?usp=sharing&oid=118058780653881782220&rtopof=true&sd=true>

5.0 CONCLUSION

The project successfully transitioned Workstations 2 and 3 from a non-functional state to a high-efficiency, networked system. By implementing modern distributor box wiring and complex PLC/Cobot handshaking, the production line is now a benchmark for industrial automation standards. This experience has provided me with deep expertise in both the mechanical and logical layers of Industrial IoT and automation.

APPENDIX



6.0 OVERVIEW

Terangkan project ni apa

7.0 OBJECTIVE

Objektif yang perlu dicapai pada project ni

8.0 TASK & RESPONSIBILITY

Terangkan setiap task yang dibuat beserta gambar kalau ada. Bagitau buat io checklist, dismantle old sensor, rewiring using new wiring technology approach(distributor box/svb), update drawing eplan, buat wiring untuk vfd dgn function code, buat plc programming utk workstation, buat plc programming utk communicate scp dgn mcp dgn cobot. Hmi punya ladder pon sama. Update plc addressing. Buat mechanical installation utk bracket dkt workstation dgn hmi dgn wireduct dan sebagainya.

9.0 TECHNICAL RESOURCES

Letak eplan report, mungkin boleh buat linkdrive and mungkin boleh attach dalam task and responsibilities. Boleh juga masukkan summary sheet jadual gerak kerja drpd google drive

10.0 CONCLUSION

Conclusion mcm biasa

APPENDIX

Letak tambahan gambar on site